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Metal-organic frameworks-derived CaO/ZnO composites as stable catalysts for biodiesel production from soybean oil at room temperature

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Biodiesel presents a sustainable alternative to fossil fuels, yet traditional homogeneous catalysts like sodium and potassium hydroxide face challenges with separation and reuse. Calcium oxide (CaO) is an effective heterogeneous catalyst for biodiesel production, but its chemical instability under reaction conditions restricts its long-term performance. This study introduces MOF-mediated synthesis (MOFMS) of heterogeneous catalysts, specifically CaO@ZnO and ZnO@CaO nanocomposites, from inexpensive and non-toxic metal salts and linkers in water. Comprehensive characterization techniques, including XRD, FT-IR, BET, FE-SEM, ICP, and CO_2-TPD, were employed to analyze these catalysts. When applied to biodiesel production from soybean oil at ambient temperature and pressure, CaO@ZnO and ZnO@CaO achieved impressive biodiesel conversion rates of 99% and 92%, respectively, within 25 min. Both catalysts maintained their activity over six utilization cycles, with Ca^{2+} leaching remaining below 4% (2% for CaO@ZnO and 4% for ZnO@CaO) after the sixth run. These results provide valuable insights into catalyst preparation and leaching control, enhancing reusability in biodiesel production. Future research should aim to improve the long-term stability and reusability of these catalysts, investigate their performance with various feedstocks, and evaluate the feasibility for industrial applications.

Keywords Calcium oxide, Mixed metal oxide, Metal-organic frameworks, Transesterification, Biodiesel production, Heterogeneous catalyst

One of the global problems in recent years is the significant increase of greenhouse gases such as CO_2, NO_x, SO_x, and CO in the atmosphere, mostly due to the excessive use of fossil fuels by humans[1,2]. Therefore, using energy sources that are non-toxic, biodegradable, and compatible with the environment has attracted attention of the scientists. Biodiesel is a fatty acid methyl ester (FAME) that is a suitable alternative to fossil fuels. It is synthesized through the transesterification of renewable sources such as soybean, palm, and vegetable oils with methanol or ethanol in the presence of a catalyst[3]. Transesterification or alcoholysis, is a process in which the alkoxy group of an ester is replaced by another alcohol group. This reaction is similar to hydrolysis, but instead of using water, an alcohol is used[4,5]. During the past years, different homogeneous and heterogeneous catalysts have been used for transesterification. Among them, homogeneous basic and acid catalysts such as NaOH, KOH, HCl, and H_2SO_4 have shown high selectivity and activity. However, they are not recoverable and may not be completely separated from the final product. While the separation of homogeneous catalysts from the products is a challenging process, heterogeneous catalysts are easy to recover and reuse[6]. In addition, enzymes have been used; however, despite their high catalytic activity, they are very expensive[7].

Heterogeneous catalysts including alkali metal oxides and mixed metal oxides such as CaO/MgO,[8] CaO/ZnO,[9] and CaO/Al_2O_3[10] have been used for this purpose. Among the metal oxides, CaO has been widely used for biodiesel synthesis due to its advantages such as availability, low cost, high basicity, and high catalytic activity. Nano-sized ZnO with high specific surface area, appropriate pore size, and high catalytic performance, also

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has been applied as a good catalyst for biodiesel production[11]. However, many of these heterogeneous catalytic systems require heating using various methods, such as oil/water bath heating, microwave irradiation, or ultrasound radiation. Therefore, there is a need to investigate more efficient and cost-effective CaO catalysts for fast transesterification reactions under room temperature conditions.

Calcium oxide can be obtained from various sources such as eggshell waste[12], shell[13], sugar factory waste[14], snail shells, and cow bones[15]. In a recent research, CaO/K_2CO_3 heterogeneous catalyst was prepared and applied for the production of biodiesel from edible oils, in which the brown algae Sargassum Oligosystem and CaO obtained from egg shells were used[16]. Ngamcharussrivichai et al.[17] have synthesized a catalyst containing a mixture of calcium/zinc oxides with different ratios of calcium and zinc, showing that the presence of calcium oxide plays a vital role in the activity of the catalyst. On the other hand, different metal oxides such as NiO-Nd_2O_3, La_2O_3, and SrO-Al_2O_3 have been used for modifying the catalytic activity of CaO[18]. Despite the potential advantages of CaO in biodiesel synthesis, the issue of Ca^{2+} leaching is still a challenge[19]. When methanol is used as the solvent, active CaO species tend to leach into the reaction medium, reducing product quality and biodiesel yield. Additionally, the reusability of CaO-based catalysts is a big challenge due to Ca^{2+} leaching or blockage of the active sites[10]. In the research conducted by Krishnamoorthy et al.[20,21], it was found that Ca^{2+} leaching from CaO nanoparticle-based catalysts was a serious problem that gradually reduced the availability of active sites for reactants. To overcome these limitations, some metal oxide mixtures or substrates, preferably porous ones, could be used to stabilize the active sites[22–24]. Metal-organic frameworks (MOFs) have recently received much attention to be used purely or post-treated as substrates to solve these limitations as well as to improve the properties of CaO[21,25].

MOFs are a new category of hybrid porous materials that are formed by connecting organic bridging ligands with metal or metal clusters[26]. The unique characteristics of MOFs such as high adsorption capacity, flexibility, porosity, numerous active sites, and high surface area lead to the application of MOFs in various fields such as gas storage, separation, drug delivery systems, and catalysts[27,28]. For example, MIL-100 (Fe) impregnated with calcium acetate was converted into a magnetic calcium oxide-based substrate by calcination, which was easily separated from the reaction mixture. The material showed good catalytic activity in biofuel synthesis at 65 °C within 2 h[29]. In another research, a zirconium fumarate MOF with unique thermal, mechanical, and catalytic properties was synthesized and used as a catalyst in biodiesel synthesis[23]. As one more example, UiO-66 MOF, with calcium acetates stabilized inside its structure by calcination, has been used as a CaO/ZrO_2 catalyst in the synthesis of biodiesel[30]. In biodiesel production, edible[19,31] and non-edible oils[32,33] have been used in the presence of different catalysts. In Iran, there are approximately 74,993 hectares of cultivated soybean that can produce 178818.19 tons production while the yield production potential of biodiesel is 34,764 tons per year[34]. Considering this fact, soybean oil has been chosen as the source of biodiesel production in this project.

In continuation of our previous research[35–37], we report the synthesis of novel, effective, and reusable catalysts for biodiesel production from soybean oil under mild conditions (room temperature, 25 min). The catalysts are synthesized using two MOFs as precursors, which are prepared from inexpensive starting materials via MOF-mediated synthesis (MOFMS) in water as a nontoxic solvent under ambient conditions. In contrast to traditional methods, which typically require the use of two MOFs to synthesize bimetallic oxides, we applied two strategies to address the issue of Ca^{2+} leaching. The resulting catalysts were evaluated for their performance in biodiesel production reactions. Incorporation of calcium acetate into the pores of ZIF-8 and zinc nitrate into a calcium-fumarate-based MOF, followed by calcination, yields CaO@ZnO and ZnO@CaO nanocomposites. Using MOFs as metal precursors provides a porous structure and a synergistic effect between the mixed metal components, resulting in accessible active sites and preventing loss of catalytic activity.

Experimental

Materials and methods

Anhydrous methanol, bulk CaO powder, $Ca(CH_3COO)_2{\cdot}xH_2O$, $Zn(NO_3)_2{\cdot}6H_2O$, sodium hydroxide, ethylene glycol, fumaric acid, and $Ca(NO_3)_2{\cdot}4H_2O$ were purchased from Merck and used without further purification. 2-Methyl imidazole (2-MeIM, 98%), triethylamine (TEA), *n*-hexane, and soybean oil were purchased from Sigma-Aldrich chemical company. Also, technical methanol was purchased from Petrochem company of Iran.

X-ray diffraction (XRD) technique was used to ensure that the catalysts were successfully synthesized. The XRD analysis, carried out on a Bruker D8 Advance instrument using Cu Kα (λ = 1.5406 Å) radiation at 2θ of 25 to 80°, indicate the crystalline nature and structure of the catalysts. Specific surface area, pore volume, and pore size were measured by N_2 adsorption/desorption at 77.0 K using a BELSORP mini II instrument. Fourier Transform Infrared spectra (FT-IR) were collected in the range 400–4000 cm^{-1} by a JASCO 6300D instrument. ^{1}H NMR (400 MHz) spectra were recorded in CD_3Cl solvent on a Bruker-Avance 400 spectrometer. Field emission scanning electron microscopy (FESEM) images were captured on a TESCAN MIRA3 instrument. The chemical Elemental concentration of the as-synthesized samples was determined with inductively coupled plasma optical emission spectrometry (ICP-OES) on a PQ9000 SERIES instrument. The basicity of the catalyst was evaluated by CO_2-temperature programmed desorption (CO_2-TPD) on a NanoSORD (made by Sensiran Co., Iran) Chemisorption Analyzer with a TCD detector. The acid number was determined using potentiometric titration with a Metrohm 848 titrator (Metrohm AG, Switzerland). The water content of soybean oil was measured using the Karl Fischer Aquamax KF method (GR Scientific, England).

The ZIF-8 and calcium-fumarate-based MOF were synthesized according to the reported procedures[38,39]. Also, the CaO nanoparticles were prepared and characterized as reported[40]. (Section S1). FAMEs in biodiesel were identified using gas chromatography-mass spectroscopy (GC-MS) analysis on an Agilent GC 6890 system equipped with a mass detector (Agilent 5973 network) and an Agilent HP-5 MS capillary GC column with internal diameter of 0.25 mm and length of 30 m were used, and temperature range with initial 80 °C (hold

2 min) to 280 °C with 5 °C per minute (hold 15 min) and total runtime 57.0 min was employed. While the rate of helium flow as carrier gas was 1 mL/min for separation. The sample injection volume was 0.5 μL.

Synthesis of CaO@ZnO nanocomposite

The catalysts used in this reaction were synthesized by the impregnation method[29]. For the synthesis of CaO@ZnO catalyst, ZIF-8 (0.9 g) was added to a solution of calcium acetate (0.31 g) in water (40 mL). The obtained mixture was stirred for 3 h at room temperature and then kept first at 120 °C for 15 h and then in a vacuum oven at 80 °C for 8 h. Finally, the sediment was calcined in a tube furnace under air atmosphere with a heating rate of 10 °C/min from 30 to 800 °C and kept at this temperature for 2 h[29]. It is noteworthy that the Ca: Zn ratio in the resulting CaO@ZnO catalyst is 1:2. Furthermore, two other CaO@ZnO composites with Ca: Zn molar ratios of 1:1 and 1:3 were synthesized according to the described method.

Synthesis of ZnO@CaO nanocomposite

For the synthesis of ZnO@CaO, calcium-fumarate-based MOF (0.4 g) was added to a solution of $Zn(NO_3)_2{\cdot}6H_2O$ (0.8 g) in water (101.3 mL). The white resulting precipitate was stirred for 3 h at room temperature. The mixture was kept at 120 °C for 15 h and then in the vacuum oven at 80 °C for 8 h. Finally, ZnO@CaO was obtained by calcination at 800 °C under the conditions described above.

Biodiesel production catalyzed by CaO@ZnO and ZnO@CaO

The transesterification reaction was carried out in a vial at room temperature in the presence of synthesized catalysts. Methanol and catalyst (5.4 wt%, 20 mg) were mixed and vigorously stirred. Then, soybean oil was added to this solution and the reaction mixture was stirred for 25 min at room temperature. The methanol to oil molar ratio was 40:1 in the reaction. At the end of the reaction, n-hexane was added to the reaction mixture, and the mixture was centrifuged at 8000 rpm for 10 min to separate the pure product. After centrifugation, two liquid layers were formed: the upper layer of hexanes contained the biodiesel product and possibly unreacted oil, while the lower layer included methanol and glycerol phase. After filtering the catalyst out, the top layer was transferred into a beaker and placed in a laboratory oven at 70 °C to evaporate the hexanes. Then, the crude biodiesel was analyzed by ^{1}H NMR spectroscopy. The conversion C_{ME} (in %) of soybean oil to biodiesel by the catalysts can be calculated from the integration values of the glyceridic and methyl ester protons in ^{1}H NMR using Eq. 1 (Scheme S1, Figures S5 and S6)[41].

$$C_{ME=100} \times \frac{5 \times I_{ME}}{5 \times I_{ME} + 9 \times I_{TAG}} \tag{1}$$

IME the integration value of the methyl ester peak.

ITAG integration value of the glyceridic peaks in the triacylglycerides (TAG) of the oil.

Recovery and reuse

The reusability of the nanocomposites in the transesterification reaction of soybean oil with methanol was evaluated as follows. At the end of each catalytic cycle, the catalyst was separated by centrifugation, washed with *n*-hexane and methanol, and activated again in the muffle furnace at 800 °C for 2 h and reused with a fresh aliquot of the substrates. The amount of Ca^{2+} leached was measured for fresh and recovered catalysts after the final cycle by ICP analysis. For comparison, the catalytic activities of nano CaO and bulk CaO catalytic activity were also investigated in the transesterification reaction. The structure of the recovered catalysts was monitored by XRD after the last cycle.

Results and discussion

Characterization of the catalysts

Two nanocomposites including CaO@ZnO and ZnO@CaO were prepared by impregnation method to reduce the Ca^{2+} leaching. The MOFMS of CaO@ZnO and ZnO@CaO were performed by stabilizing calcium acetate inside the ZIF-8 structure and zinc nitrate in calcium-fumarate-based MOF, respectively; which is then followed by calcination (Fig. 1). Both of the precursors, CaO and ZnO, possess basic sites with different levels of basicity for the transesterification reaction, although they suffer from Ca^{2+} leaching and make reusability problematic.

The XRD patterns of the bulk CaO (**a**), nano CaO (**b**), CaO@ZnO (**c**), ZnO derived from ZIF-8 (**d**) and ZnO@CaO are displayed in Fig. 2. The XRD pattern for the CaO sample shows strong and sharp peaks at (2θ = 32° (111), 37° (200), 54° (220), 64° (311) and 67° (222)) (ICDD PDF no.: 01-074-1226) that confirm the presence of a cubic form of the Fm-3 m type structure of CaO as the main phase with good crystallinity. While the other diffraction peaks at (2θ = 32° (100), 34° (002), 36° (101), 47° (102), 57° (110), 63° (103), 67° (200), 68° (112) and 69° (201)) (ICDD PDF no.: 01-075-1526) correspond to hexagonal form of the P63mc type structure of ZnO. This information confirmed the presence of CaO and ZnO phases in both nanocomposites.

The N_2 adsorption/desorption isotherms for studying the porosity of the catalysts are presented in Fig. 3; Table 1. Bulk CaO has a specific surface area of 7 m^2/g while nano CaO has a specific surface area of 50 m^2/g. The specific surface areas of the nanocomposites CaO@ZnO and ZnO@CaO are 12 and 10 m^2/g, respectively

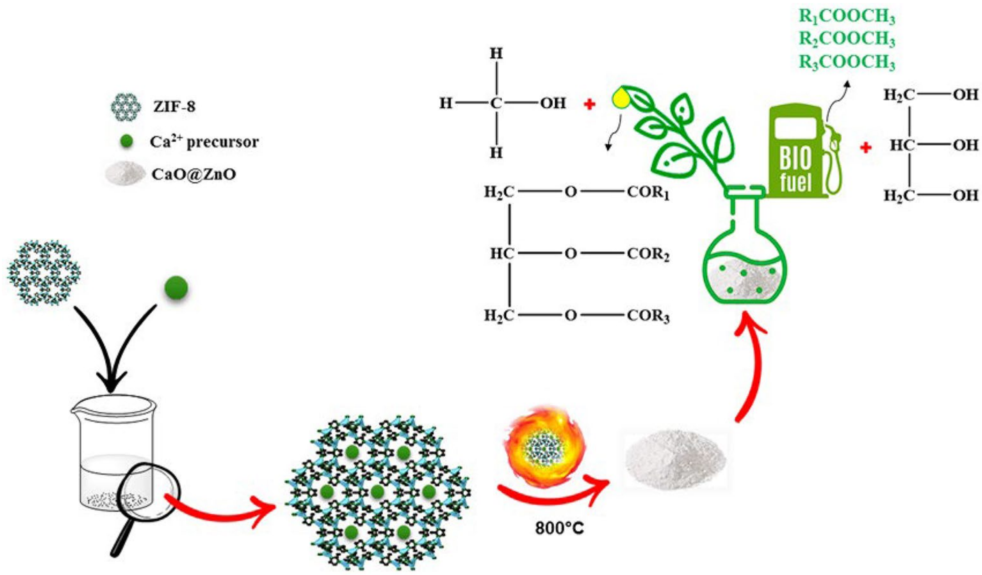

Fig. 1. Schematic illustration of the synthesis of CaO@ZnO catalyst.

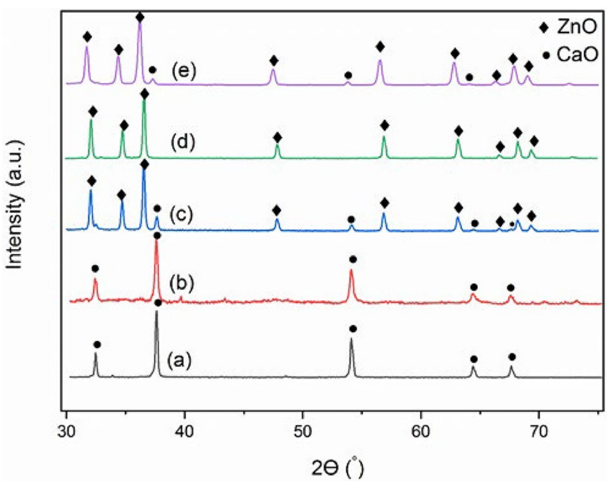

Fig. 2. XRD patterns of the bulk CaO (**a**), nano CaO (**b**), CaO@ZnO (**c**), ZnO derived from ZIF-8 (**d**), and ZnO@CaO (**e**) materials.

Sample	S_{BET} ($m^2\,g^{-1}$)	Total pore Volume (cm^3g^{-1})	Mean pore diameter (nm)
bulk CaO	7	0.053	32
nano CaO	50	0.22	18
ZnO derived from ZIF-8	8	0.02	14
CaO@ZnO	12	0.040	13
ZnO@CaO	10	0.1	50

Table 1. BET data related to bulk CaO, nano CaO, ZnO derived from ZIF-8, CaO@ZnO and ZnO@CaO catalysts.

(Table 1). The decreased S_{BET} of CaO@ZnO is attributed to the CaO and ZnO bearing larger crystallite sizes at a higher temperature. In general, the calcined.

samples at 800 °C have lower BET surface areas compared to nano CaO that calcined at 500 °C[42]. The pore volume of bulk CaO, nano CaO, CaO@ZnO, and ZnO@CaO were 0.053, 0.22, 0.04, and 0.1 cm^3/g, respectively. The CaO is accessible to sintering at a higher temperature, thus resulting in CaO grain growing up and squeezing

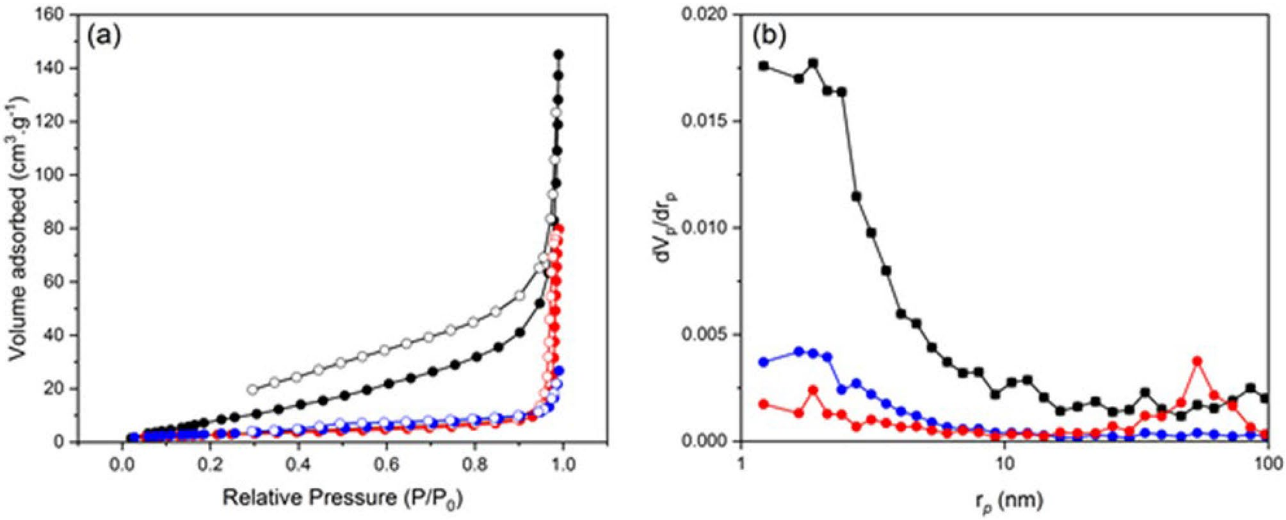

Fig. 3. N2 adsorption/ desorption isotherms of nano CaO (black), CaO@ZnO (red), and ZnO@CaO (blue) (**a**), and the corresponding BJH (**b**).

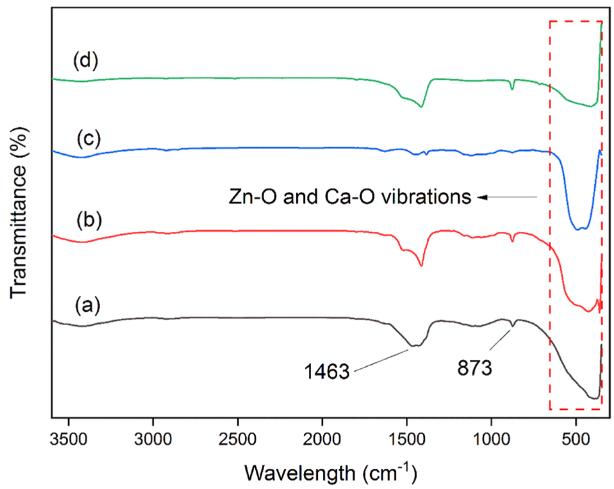

Fig. 4. FT-IR spectra of bulk CaO (**a**), CaO@ZnO (**b**), ZnO derived from ZIF-8 (**c**), and ZnO@CaO (**d**).

Temperature range (°C)	Basic sites (μmol.g^{-1})
[43.3–360]	15.5
[360–665]	463
[665-804.5]	848.4

Table 2. Basic strength and basic sites of CaO@ZnO catalyst measured by CO2-TPD.

the porous space[43,44]. Wang et al.[44] reported that CaO-based grains would agglomerate and grow up leading to the blockage of pore structure, where the micro pore space was first squeezed during this process.

The FT-IR spectra of the bulk CaO (a), CaO@ZnO (b), ZnO derived from ZIF-8 (c), and ZnO@CaO (d) are shown in Fig. 4. According to the literature, the absorption bands of Zn-O and Ca–O appeared at 400–600 cm^{-1}.[45,46] The bands at 873 and 1463 cm^{-1} in the CaO spectrum (a) are attributed to C–O bond of carbonate (CO_3^{-2}) group[47]. The carbonate bands indicate that the sample was contaminated by atmospheric CO_2[48].

The total number of basic sites plays a crucial role in influencing the catalytic efficiency of the transesterification catalysts. Figure 5; Table 2 show the case CO_2-TPD profiles and basic site quantities of the CaO@ZnO catalyst, respectively. Classification of the basic sites, the types and strengths, using CO_2-TPD spectra was segmented into three categories: weak basic site (< 400 °C), medium basic site (400–550 °C), and strong basic site (> 550 °C).[49]

The higher temperature of CO_2 desorption could be related to the higher basicity and basic strength of CaO@ZnO (Fig. 5).

The morphologies of CaO@ZnO and ZnO@CaO were studied by FE-SEM images. The SEM images of these catalysts, the corresponding particle size distribution histograms, and dispersive X-ray (EDX) mapping analysis are presented in Fig. 6. Both CaO@ZnO and ZnO@CaO composites showed a uniform agglomerated structure with sintered nanoparticles. The average particle size of about 74 nm for CaO@ZnO and 69 nm for ZnO@CaO were calculated from the FE-SEM images. The EDX analysis and mapping were used to detect the elements in the catalysts, which indicate that the distribution of Ca, O, and Zn is uniform in the CaO@ZnO composite (Figure S4). Similarly, the distribution of these elements is homogeneous in ZnO@CaO, as depicted in Fig. 6h and j-l. The advantage of MOF derivatives, in allowing orderly distribution of the elements, is further confirmed. This homogeneous distribution promotes enhanced transesterification conversion as it facilitates full contact between the reactants and the active sites.

Biodiesel production

Here, we focused on the optimization of the transesterification reaction using the synthesized composites. Since the nature of both composites is the same, the CaO@ZnO composite was applied to optimize the biodiesel production reaction conditions. In this regard, different factors including methanol to soybean oil ratio, reaction time, and catalyst amount were carefully investigated. Additionally, the effect of the Ca/Zn molar ratio on the performance of the catalysts was evaluated.

Effect of methanol to oil ratio

The effect of methanol to oil molar ratio (25:1, 35:1, 40:1, 50:1, 60:1, and 70:1) on biodiesel production was investigated. The highest conversion was achieved with the methanol to oil ratio of 40:1 at mild conditions. As shown in Fig. 7, the conversion dramatically increased by increasing the methanol to oil ratio. The technical methanol from Petrochem company in Iran was used instead of anhydrous methanol and CaO@ZnO was investigated under optimized conditions. ^{1}H NMR spectrum of produced biodiesel demonstrated 99% yield which is the same as the results obtained using anhydrous methanol.

Effect of reaction time on biodiesel production efficiency

To investigate the effect of time on biodiesel production, five mixtures containing 20 mg CaO@ZnO, 0.37 g soybean oil, and 2 g methanol were prepared and stirred for 5, 10, 15, 20, and 25 min at room temperature. The yield of the biodiesel was determined by ^{1}H NMR spectroscopy. The results shown in Fig. 8 indicate that by increasing the reaction time, the reaction efficiency increased from 35 to 99% for CaO@ZnO. Accordingly, 25 min was chosen as the optimal reaction time.

Additionally, the catalytic activity of the synthesized composites, CaO@ZnO and ZnO@CaO, was compared to that of nano and bulk CaO. The investigation was conducted under identical conditions ranging from 5 to 60 min. The results presented in Fig. 8 demonstrate that nano CaO exhibits superior performance compared to bulk CaO on catalytic activity, confirming the positive influence of a slightly higher surface area of nano CaO compared to bulk CaO[50]. The synthesized composites exhibit excellent activity, surpassing both nano and bulk CaO. Notably, CaO@ZnO demonstrates the highest biodiesel yield among the tested catalysts in 25 min. Furthermore, both CaO@ZnO and ZnO@CaO composites exhibit similar efficiency when the reaction time is extended from 25 to 60 min.

Optimization of the catalyst amount

The amount of catalyst is an important factor in the effective production of biodiesel. In this research, 5, 10, 15, 20, and 25 mg (1.3, 2.7, 4, 5.4 and 6.7 wt%, respectively) of the catalyst were selected. The results indicated that

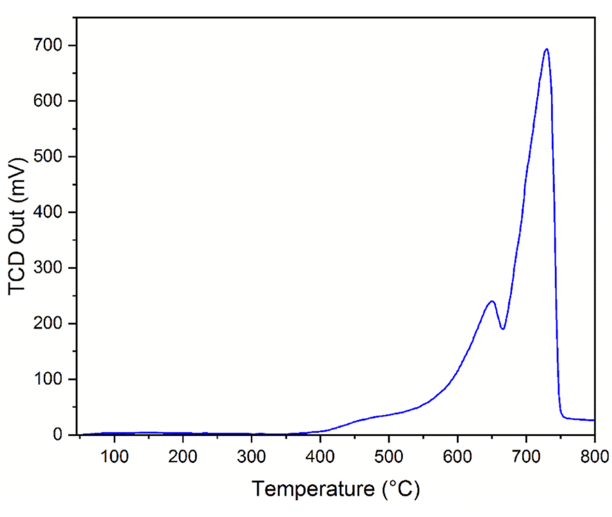

Fig. 5. CO_2-TPD of CaO@ZnO.

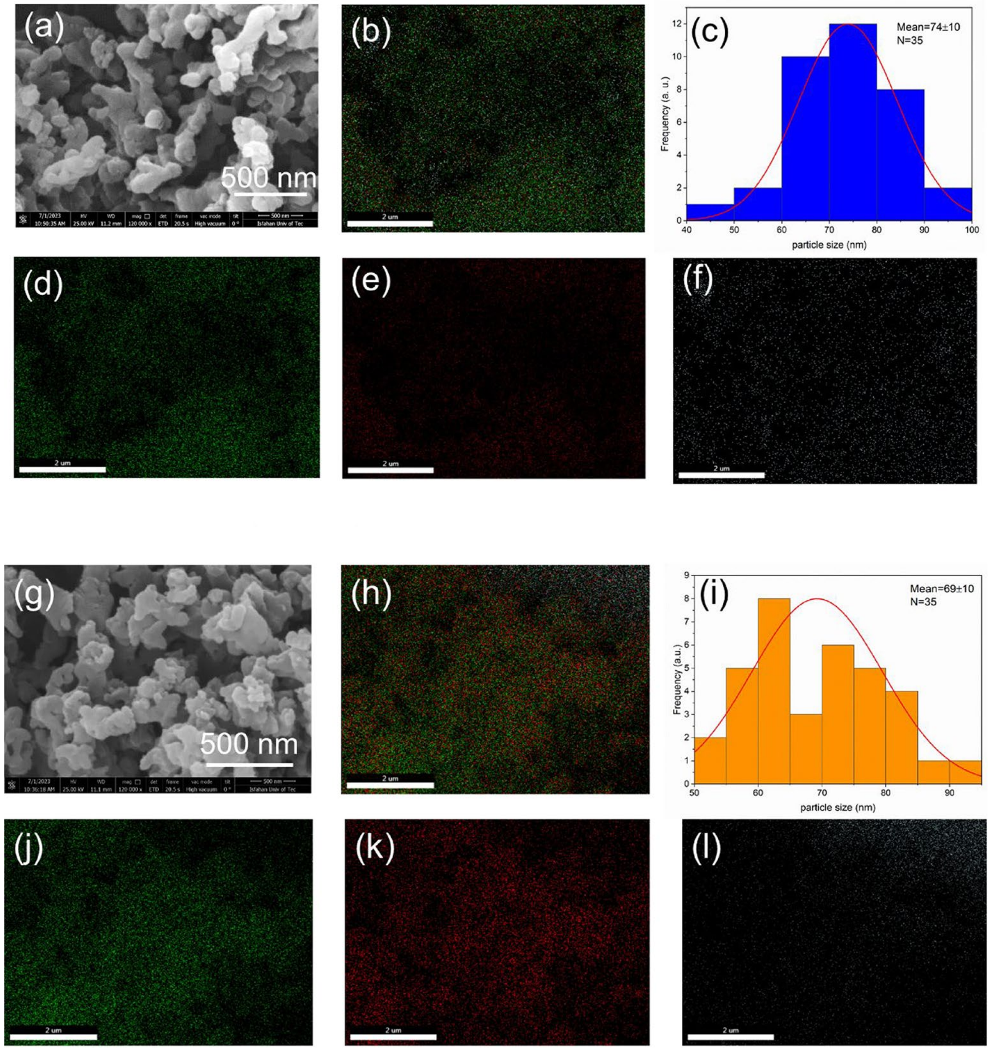

Fig. 6. FE-SEM images (**a**, **g**), EDX mapping (**b**, **h**), particle size distribution histograms (**c**, **i**) and element distribution of Ca (**d**, **j**), O (**e**, **k**), and Zn (**f**, **l**) for CaO@ZnO and ZnO@CaO, respectively.

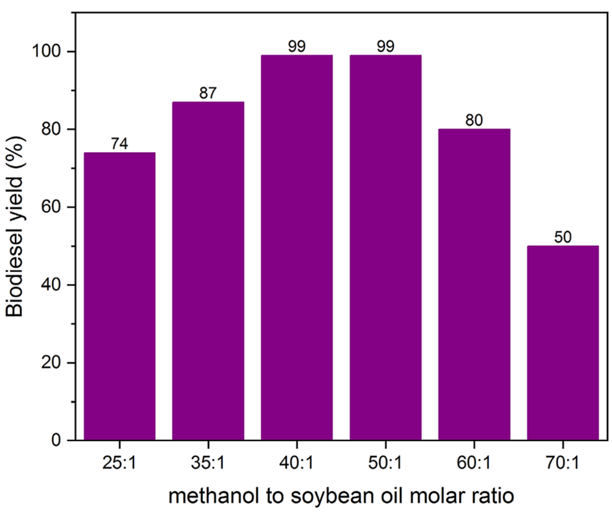

Fig. 7. Effect of the amount of methanol to oil on transesterification of soybean oil with methanol. (reaction conditions: a mixture of methanol and oil; 20 mg CaO@ZnO composite; room temperature; 25 min).

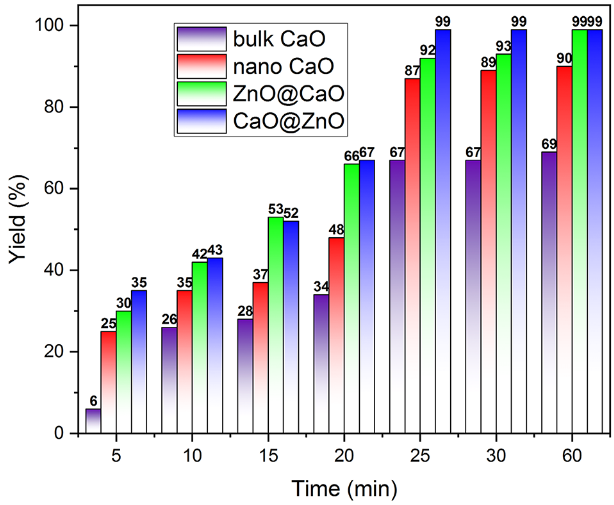

Fig. 8. Biodiesel production at different reaction times using 20 mg catalyst. (Reaction conditions: 0.37 g soybean oil; 2 g methanol; 20 mg catalyst; room temperature).

Catalyst	Ca/Zn molar ratio	Yield (%)
CaO@ZnO	1:1	70
CaO@ZnO	1:2	99
CaO@ZnO	1:3	30
CaO@ZnO	1:0.5	80

Table 3. Catalytic performance at different Ca/Zn molar ratios.

the yield of the desired ester was 65%, 71%, 84%, 99%, and 99%, respectively. Accordingly, 20 mg of catalyst (5.4 wt%) was chosen as the optimal amount. The results are summarized in Fig. 9.

Optimization of Ca/Zn molar ratio
The effect of different molar ratios of Ca/Zn (1:0.5, 1:1, 1:2, and 1:3) in the CaO@ZnO composite, was investigated using the same reaction conditions. The results presented in Table 3 indicate that the molar ratio of 1:2 provides the highest yield, which was selected as the optimum for the CaO@ZnO catalyst synthesis. The results show that encapsulation of CaO precursor in ZIF-8 enhanced the catalytic activity of Zn-MOF[51]. However, the presence of ZnO was crucial in increasing the dispersion of CaO and reducing its agglomeration during the reaction[52].

The chromatogram of produced biodiesel was used for the determination of the FAME obtained by CaO@ZnO on the optimized reaction conditions (Figure S7 and Table S1) The profile of the FAME indicates that the produced biodiesel contains mainly 9,12-octadecadienoic acid methyl ester (37.82%), hexadecanoic acid methyl ester (18.17%) and octadecanoic acid methyl ester (9.96%).

Reusability
The unique advantage of heterogeneous catalysts, in comparison to homogeneous catalysts, is their reusability. The reusability of CaO@ZnO and ZnO@CaO was evaluated in the biodiesel production process and the results were compared with bulk and nano CaO. After each run, the catalysts were washed and activated in a muffle furnace at 800 °C for 2 h to remove any remaining oil or biodiesel, adsorbed moisture, and CO_2 molecules from the catalyst surface[53]. Then, the biodiesel production was carried out following the procedure described in the experimental section. From the results shown in Fig. 10, it can be inferred that the oil conversion obtained from CaO@ZnO and ZnO@CaO decreased from 99 to 83% and 92–73%, respectively, after the fourth cycle, ending at 70% and 65%, respectively after the sixth cycle. In contrast, a rapid decline from 88 to 45% for nano CaO and 67–37% for bulk CaO was observed after the fourth cycle (Fig. 10a). The deactivation of the catalysts may be attributed to the hydration of CaO by H_2O and the reaction with CO_2 from air during the separation processes. The decreased catalytic efficiency observed in the later cycles can be attributed to the potential blockage of the active sites[48]. Additionally to explore the effect of catalyst activation, recovered CaO@ZnO catalyst has been used

without activation and the results are shown in the Fig. 10b. Two main factors affect biodiesel production in this kind of catalyst. The first one is the surface area of the catalyst which results in accessible active sites for reaction. The second one is the type of active site which relates to the type of metal. As shown in Table 1, the specific surface area of bulk CaO (7 m^2g^{-1}) is considerably smaller than the surface area of nano CaO (50 m^2g^{-1}), which leads to higher biodiesel production (the first factor). On the other hand, in comparison of nano CaO with

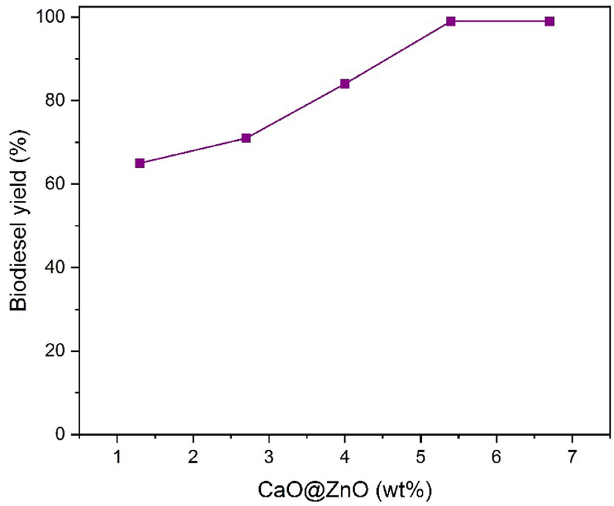

Fig. 9. Effect of CaO@ZnO amount on FAME conversion. (Reaction conditions: 0.37 g soybean oil, 2 g methanol, room temperature, 25 min).

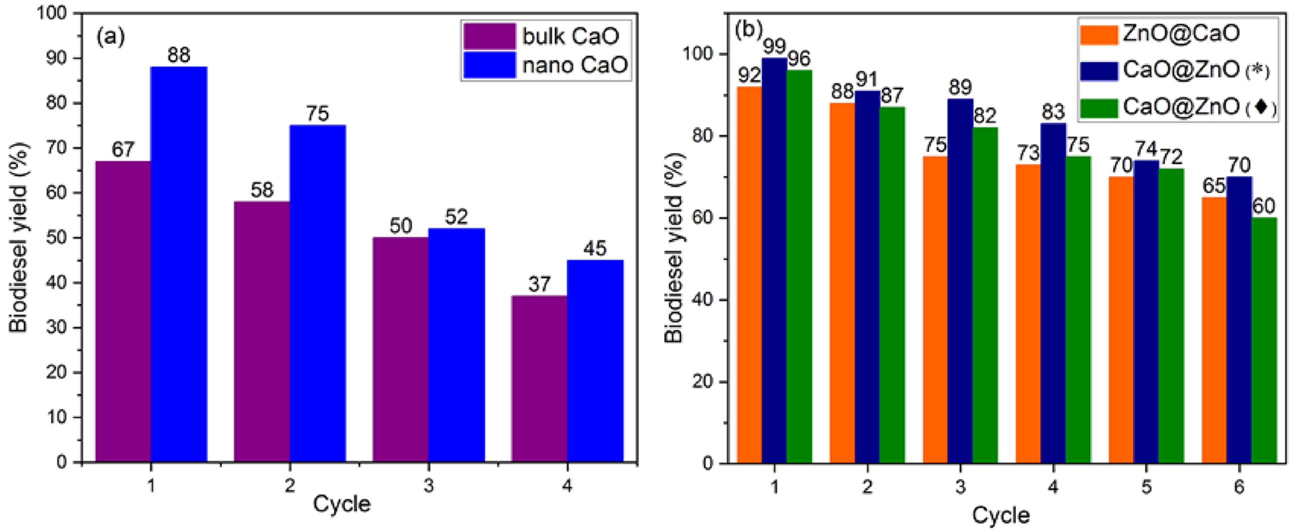

Fig. 10. Reusability of bulk CaO and nano CaO (**a**), ZnO@CaO and CaO@ZnO (**b**). (*calcination after each run ⬩without calcination after each run)

the prepared nanocomposites the type of active site is more important and the results reveal the mixed metal oxides have a synergistic effect and therefore, increase the yield of biodiesel[22].

The structural stability of the recovered catalysts was monitored by XRD (Fig. 10). The obtained XRD pattern from the recovered catalyst after six cycles shows that the catalysts retained their structure.

To investigate the leaching amounts of Ca^{2+}, the calcium concentrations of the fresh and recycled catalysts were measured using the ICP method. The results are presented in Table 4. The percentages of leached Ca^{2+} for bulk CaO, nano CaO, ZnO@CaO, and CaO@ZnO were 32.5%, 14%, 4%, and 2%, respectively. Using mixed metal oxide overcomes to leaching problem and results in excellent reusability of ZnO@CaO and CaO@ZnO compared to the conventional oxides[22].

During transesterification, the leached Ca^{2+} species can react with free fatty acids (FFAs) and form soap[54]. To check this point, both biodiesel and glycerol phases were analyzed to measure the amount of leached Ca^{2+}. The results are summarized in Table 5. As can be seen, the total amounts of leached Ca^{2+} (in biodiesel and glycerol phases) and the amount of Ca^{2+} in the recovered catalyst are equal to the amount of Ca^{2+} in the fresh catalyst. On the other hand, the acid number of soybean oil was obtained at about 0.58 mg KOH/g which is less than 2.0 mg KOH/g. Also, the water content of the oil was found to be 0.08 wt%. Consequently, this little amount of water reduces the saponification process during the transesterification.

The efficiencies of CaO@ZnO and ZnO@CaO were also compared with some of the previously reported catalysts. As can be seen in Table 6, this method is superior to the previously reported catalysts in terms of

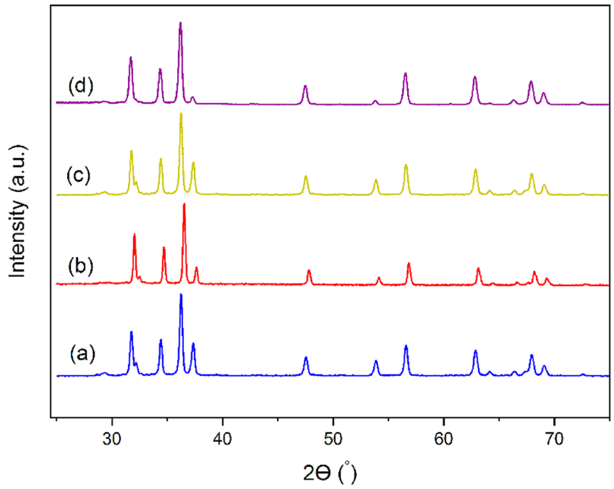

Fig. 11. XRD patterns of CaO@ZnO (**a**), recovered CaO@ZnO (**b**), ZnO@CaO (**c**), and recovered ZnO@CaO (**d**).

Catalyst	No. of cycles	Ca^{2+}amount[a]		Leached Ca^{2+}(%)
		Fresh catalyst	Recovered catalyst	
Bulk CaO	4	66.36	33.86	32.5
Nano CaO	4	36.29	22.23	14
ZnO@CaO	6	15.63	11.63	4
CaO@ZnO	6	43.52	41.50	2

Table 4. The amount of Ca^{2+} concentration in fresh and reused catalysts. [a]amount of Ca^{+2} (wt%)

Catalyst or phase	Leaching Ca^{+2}
CaO@ZnO (before transesterification)	4.65[a] (43.52)[b]
CaO@ZnO (after transesterification)	4.59 [a] (42.89)[b]
Glycerol	0.039[a]
Biodiesel	0.02 [a]

Table 5. The analysis of glycerol and biodiesel phases for determination of Ca2+ leaching. [a]amount of Ca^{+2} (ppm) [b]amount of Ca^{+2} (wt%)

Entry	Catalyst	Feedstock	C[a] (wt%)	M/O[b]	t[c] (min)	T[d](°C)	FAME (%)	Reusability, FAME (%)	Ref.
1	CaO/Fe_2O_3	palm oil	4	9	120	65	95.07	5th, 62.51	[29]
2	CaO/zeolite	soybean oil	30	9	180	65	95	-	[55]
3	$MgO-CaO/SiO_2$	repeseed oil	3.5	6.5	210	68	98	3th, 88.7	[56]
4	$MgFe_2O_4$@CaO	soybean oil	1	12	180	70	98.03	5th, 89	[57]
5	ZnCaO	waste cooking sunflower oil	5	14	120	57.5	96.5	-	[58]
6	Eggshell derived CaO catalyst	Argemone mexicana oil	3.05	9.7	180	60	99.07	6th, 65.75	[59]
7	CaO@ZnO	soybean oil	5.4	40	25	25	99	6th, 70	This work
8	ZnO@CaO	soybean oil	5.4	40	25	25	92	6th, 65	This work

Table 6. Comparison of catalytic performance of different catalysts in biodiesel preparation reaction. [a] C: amount of catalyst. [b] M/O: Methanol to oil molar ratio. [c] t: reaction time. [d] T: reaction temperature.

reaction temperature, time, and reusability. CaO loading on the other supports including Fe_2O_3, Zeolite, and SiO_2 (Table 6, entries 1–3) has been reported and the results displayed higher temperatures and longer time toward the synthesized catalysts (Table 6, entries 7 and 8). Compared to mixed metal catalysts (Table 6, entries 4 and5), obtained nanocomposites (Table 6, entries 7 and8) achieved high-yield biodiesel at a lower temperature.

Conclusion

In this research, the reusability and catalytic activity of CaO were efficiently improved by successfully preparing CaO/ZnO nanocomposites via simple calcination of the precursor hybrid materials containing MOFs. The precursors were prepared easily by one-pot routes through the impregnation of calcium and zinc salts in inexpensive zinc and calcium MOFs, respectively. The CaO@ZnO and ZnO@CaO composites exhibit excellent catalytic activity in the transesterification of soybean oil to biodiesel with low Ca^{2+} leaching. CaO@ZnO exhibits a maximum conversion of 99% with 5.4 wt% of the catalyst at room temperature in 25 min. The CaO@ZnO catalyst achieved 70% conversion after six cycles, with only 2% of Ca^{2+} leaching which confirms the effect of the composite structure in preventing Ca^{2+} leaching. The catalyst is promising for application in transesterification reactions, contributing to the production of environmentally friendly and sustainable biodiesel.

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

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Declarations

Competing interests
The authors declare no competing interests.

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